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« In the name of God »

نوشته شده

1) توان هر استیلا را $box - locations$ و $player - location$ در نظری می‌بینیم.
 بدین‌ترین تعریف آسان است، زیرا در آن متغیرهای وجود دارد که تغییر نمی‌کنند و حافظه‌ی بدین
 برای بهبود دانسیته متغیرهای ثابت را حذف می‌کنیم.

2) هر Action می‌تواند به موقعیت جدید و بازیکن تاسیس دارد یا پس:

Action = [up, down, right, Left]

3) environment = [H, ..., w, B, G, ...] Action = [up, down, right, Left]

observation: Box: در Goal قرار دارد یا خیر، موقعیت جدید ها و بازیکن

performance measure: الوریتم‌های DFS و BFS و IDS به علت اینکه جستجوی
 ناگهانه هستند این قابلیت را ندارند اما در A* تابع heuristic این را تعیین می‌کند.

4) اگر در اینجور مسائل بالاخص در جستجوی ناگهانه تمام حالات را همواره Expand کنیم مشکلات
 متعددی رخ می‌دهد مثلاً: I) ممکن است در مسئله‌ای بی‌پایان و هرگز به جواب نرسیم.
 II) از لحاظ زمانی کاملاً دچار مشکل شده و ممکن است حل مسئله از عمر جهان بیشتر طول بکشد.
 III) بریل استفاده زیاد از حافظه به مرور مربوط به حافظه بدین فراهم.

راه حل: با تعریف یک Set به نام visited می‌توانیم این قضیه را مدیریت کنیم. به طوری که همواره با توالی
 یک state جدید اگر در visited نبود آن را به Fringe اضافه می‌کنیم.

5) الوریتم BFS یا جستجوی سطحی بدین گونه است که هر بار یک State را Expand می‌کنیم و

تمام چه‌های آن را بررسی می‌کنیم و به چه‌هایی که Expand نشده‌اند به توالی اضافه می‌کنیم و همواره جواب بهینه را

پیدا می‌کنیم اما حافظه زیادی می‌برد و از لحاظ زمان نسبت به DFS سریع‌تر است.

اما الگوریتم DFS را جستجوی عمقی میگویند. حافظه کمتری احتیاج دارد و در زمان کمتری جواب را می یابد اما محواره جواب بهینه را تولید نمی کند. اما در بعضی از موارد و مسائل ساده سریع تر به جواب می تواند برسد اما نه جواب بهینه!

(a) جواب بهینه تولید کردن یکی از مسائل های DFS است و همچنین زمان، اما در بعضی مسائل صرف رسیدن به جواب کافی باشد و به جواب بهینه نیاز نباشد. حتی با مسئله ساده باشد و سریع تر به جواب برسد. در این مواقع و هنوز از DFS استفاده می شود.

(b) IDS با گسترش پهنای DFS باعث می شود تا جواب بهینه را بیابد و همچنین از قابلیت DFS که حافظه کمتری است استفاده کنونی از لحاظ زمانی مانند خود DFS بهینه نیست.

(6) به اساس تزارش سوال و نکته های بالا توضیح می شود.

(7) اولین هیدروستیک بر اساس فاصله box تا goal ها (فاصله منتهی) و فاصله بازگشت از جعبه ها را در نظر می گیریم اما این عام map ها با شکل نبود زیرا عقده هم پاره ها را در نظر نمی گیریم حال برای آنکه تابع heuristic یا Optimal شود فاصله box تا عقده فعلاً فاصله این دو نیست باقی حرکت از یک پرتال بهینه تر است مثلاً فاصله باکس تا پرتال a در 1 و بعد از پرتال a حتی 2 تا عقده بهینه تر باشد و خاص 2 پرتال داریم که رفتن از پرتال a در a به پرتال ط و از پرتال ط به عقده بهینه تری شود و ما هر بار می بینیم این فاصله ها را در نظر می گیریم و جمع می کنیم و فاصله بازگشت تا جعبه ها را هم می بینیم آن را جمع می کنیم. هیدروستیک جدید در زمان بهینه ای به جواب می رساند ما را.

(8) در حالت اول که پرتال ها را در نظر نمی گیریم برای نقشه های که پرتال نداشته و بعضی نقشه ها که حداقل یک پرتال داشته باشند به صورت بهینه به جواب می رسید اما بعضی مانند Map 7 به جواب نمی رسیدند.

در هیدروستیک درست سطر عام Map ها در زمان درستی به جواب رسیدند الی Map 9.

Reports

9

BFS:

```
BFS took 0.0 seconds on map 1 and visited 47 states.  
7 moves were used: UDDURLL  
BFS took 0.0 seconds on map 2 and visited 26 states.  
6 moves were used: LDUUDL  
BFS took 0.0 seconds on map 3 and visited 129 states.  
13 moves were used: ULDDUUUURDDDD  
BFS took 0.0 seconds on map 4 and visited 0 states.  
No Solution Found!  
BFS took 0.07 seconds on map 5 and visited 7842 states.  
15 moves were used: ULDDRDLLL UURUL  
BFS took 0.22 seconds on map 6 and visited 22672 states.  
34 moves were used: URUUUURLLLLLLDDDRDDDDDDLRRRRRRR  
BFS took 7.52 seconds on map 7 and visited 795574 states.  
34 moves were used: RURRDDDLD RUUUULLLRDRDRDLDLLURLU  
BFS took 0.08 seconds on map 8 and visited 11959 states.  
14 moves were used: UURDLRRDRURDR  
for map 9, we have big time to find solution.....  
BFS took 2.2 seconds on map 10 and visited 254850 states.  
46 moves were used: RRRRRDRULURULLLULDRUUULDRDLRRDRULURUDDRDLLLL
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```
BFS took 0.0 seconds on map 1 and visited 47 states.  
7 moves were used: UDDURLL  
BFS took 0.0 seconds on map 2 and visited 26 states.  
6 moves were used: LDUUDL  
BFS took 0.0 seconds on map 3 and visited 129 states.  
13 moves were used: ULDDUUUURDDDD  
BFS took 0.0 seconds on map 4 and visited 0 states.  
No Solution Found!  
BFS took 0.06 seconds on map 5 and visited 7842 states.  
15 moves were used: ULDDRDLLL UURUL  
BFS took 0.17 seconds on map 6 and visited 22672 states.  
34 moves were used: URUUUURLLLLLLDDDRDDDDDDLRRRRRRR  
BFS took 7.11 seconds on map 7 and visited 795574 states.  
34 moves were used: RURRDDDLD RUUUULLLRDRDRDLDLLURLU  
BFS took 0.07 seconds on map 8 and visited 11959 states.  
14 moves were used: UURDLRRDRURDR  
for map 9, we have big time to find solution.....  
BFS took 2.22 seconds on map 10 and visited 254850 states.  
46 moves were used: RRRRRDRULURULLLULDRUUULDRDLRRDRULURUDDRDLLLL
```

```
BFS took 0.0 seconds on map 1 and visited 47 states.  
7 moves were used: UDDURLL  
BFS took 0.0 seconds on map 2 and visited 26 states.  
6 moves were used: LDUUDL  
BFS took 0.0 seconds on map 3 and visited 129 states.  
13 moves were used: ULDDUUUURDDDD  
BFS took 0.0 seconds on map 4 and visited 0 states.  
No Solution Found!  
BFS took 0.05 seconds on map 5 and visited 7842 states.  
15 moves were used: ULDDRDLLL UURUL  
BFS took 0.21 seconds on map 6 and visited 22672 states.  
34 moves were used: URUUUURLLLLLLDDDRDDDDDDLRRRRRRR  
BFS took 7.51 seconds on map 7 and visited 795574 states.  
34 moves were used: RURRDDDLD RUUUULLLRDRDRDLDLLURLU  
BFS took 0.07 seconds on map 8 and visited 11959 states.  
14 moves were used: UURDLRRDRURDR  
for map 9, we have big time to find solution.....  
BFS took 2.24 seconds on map 10 and visited 254850 states.  
46 moves were used: RRRRRDRULURULLLULDRUUULDRDLRRDRULURUDDRDLLLL
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DFS:

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DFS took 0.0 seconds on map 1 and visited 8 states.  
7 moves were used: DURLLRU  
DFS took 0.0 seconds on map 2 and visited 9 states.  
6 moves were used: LUDLRD  
DFS took 0.0 seconds on map 3 and visited 25 states.  
23 moves were used: ULDUUURDULDDDDUUURDDDD  
DFS took 0.0 seconds on map 4 and visited 0 states.  
No Solution Found!  
DFS took 0.01 seconds on map 5 and visited 1225 states.  
131 moves were used: DLLULRURULLLDDRRDRRULRDLDLUURURDDLDLL  
DFS took 0.02 seconds on map 6 and visited 1480 states.  
513 moves were used: DRDLLDDLDRRUURRRRDRUULLURRULLLUUURDRU  
for map 7, we have big time to find solution.....  
DFS took 0.0 seconds on map 8 and visited 348 states.  
261 moves were used: URLUURRDDUULDLDRRRDDLLULLULDLDRDDLULL  
for map 9, we have big time to find solution.....  
DFS took 4.69 seconds on map 10 and visited 414948 states.  
5977 moves were used: LLDDRRURDRURDRRRUULDRRULUULLLDLURRDD
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```
DFS took 0.0 seconds on map 1 and visited 8 states.  
7 moves were used: DURLLRU  
DFS took 0.0 seconds on map 2 and visited 9 states.  
6 moves were used: LUDLRD  
DFS took 0.0 seconds on map 3 and visited 25 states.  
23 moves were used: ULDUUURDULDDDDUUURDDDD  
DFS took 0.0 seconds on map 4 and visited 0 states.  
No Solution Found!  
DFS took 0.01 seconds on map 5 and visited 1225 states.  
131 moves were used: DLLULRURULLLDDRRDRRULRDLDLUURURDDLDLLU  
DFS took 0.01 seconds on map 6 and visited 1480 states.  
513 moves were used: DRDLLDDLDRRUURRRRDRUULLURRULLLUUURDRUR  
for map 7, we have big time to find solution.....  
DFS took 0.01 seconds on map 8 and visited 348 states.  
261 moves were used: URLUURRDDUULDLDRRRDDLLULLULDLDRDDLULLU  
for map 9, we have big time to find solution.....  
DFS took 5.08 seconds on map 10 and visited 414948 states.  
5977 moves were used: LLDDRRURDRURDRRRUULDRRULUULLLDLURRDDRR
```

```
DFS took 0.0 seconds on map 1 and visited 8 states.  
7 moves were used: DURLLRU  
DFS took 0.0 seconds on map 2 and visited 9 states.  
6 moves were used: LUDLRD  
DFS took 0.0 seconds on map 3 and visited 25 states.  
23 moves were used: ULDUUURDULDDDDUUURDDDD  
DFS took 0.0 seconds on map 4 and visited 0 states.  
No Solution Found!  
DFS took 0.01 seconds on map 5 and visited 1225 states.  
131 moves were used: DLLULRURULLLDDRRDRRULRDLDLUURURDDLDLL  
DFS took 0.02 seconds on map 6 and visited 1480 states.  
513 moves were used: DRDLLDDLDRRUURRRRDRUULLURRULLLUUURDRU  
for map 7, we have big time to find solution.....  
DFS took 0.0 seconds on map 8 and visited 348 states.  
261 moves were used: URLUURRDDUULDLDRRRDDLLULLULDLDRDDLULL  
for map 9, we have big time to find solution.....  
DFS took 5.65 seconds on map 10 and visited 414948 states.  
5977 moves were used: LLDDRRURDRURDRRRUULDRRULUULLLDLURRDDRR
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IDS:

IDS took 0.0 seconds on map 1 and visited 15 states.
7 moves were used: DURLLRU
IDS took 0.0 seconds on map 2 and visited 10 states.
6 moves were used: LUDLRD
IDS took 0.0 seconds on map 3 and visited 84 states.
13 moves were used: ULDDUUUURDDDD
IDS took 0.0 seconds on map 4 and visited 0 states.
No Solution Found!
IDS took 0.22 seconds on map 5 and visited 6179 states.
15 moves were used: LULDDRDLUURUL
IDS took 2.59 seconds on map 6 and visited 7139 states.
34 moves were used: DRDDDRLLLLLLLLUUUUUUUUULRRRRRRR
IDS took 48.99 seconds on map 7 and visited 749203 states.
34 moves were used: RURRDDDLRUUUULLLRDRDRDLDLLUUDR
IDS took 0.24 seconds on map 8 and visited 6589 states.
14 moves were used: UURDLRRDRURDR
for map 9, we have big time to find solution.....
IDS took 91.13 seconds on map 10 and visited 236933 states.
46 moves were used: RRRRRDRULURULLLDLLURUULDRDLRRDRULURURDDRDLLLL

IDS took 0.0 seconds on map 1 and visited 15 states.
7 moves were used: DURLLRU
IDS took 0.0 seconds on map 2 and visited 10 states.
6 moves were used: LUDLRD
IDS took 0.0 seconds on map 3 and visited 84 states.
13 moves were used: ULDDUUUURDDDD
IDS took 0.0 seconds on map 4 and visited 0 states.
No Solution Found!
IDS took 0.21 seconds on map 5 and visited 6179 states.
15 moves were used: LULDDRDLUURUL
IDS took 2.7 seconds on map 6 and visited 7139 states.
34 moves were used: DRDDDRLLLLLLLLUUUUUUUUULRRRRRRR
IDS took 51.95 seconds on map 7 and visited 749203 states.
34 moves were used: RURRDDDLRUUUULLLRDRDRDLDLLUUDR
IDS took 0.25 seconds on map 8 and visited 6589 states.
14 moves were used: UURDLRRDRURDR
for map 9, we have big time to find solution.....
IDS took 86.6 seconds on map 10 and visited 236933 states.
46 moves were used: RRRRRDRULURULLLDLLURUULDRDLRRDRULURURDDRDLLLL

IDS took 0.0 seconds on map 1 and visited 15 states.
7 moves were used: DURLLRU
IDS took 0.0 seconds on map 2 and visited 10 states.
6 moves were used: LUDLRD
IDS took 0.0 seconds on map 3 and visited 84 states.
13 moves were used: ULDDUUUURDDDD
IDS took 0.0 seconds on map 4 and visited 0 states.
No Solution Found!
IDS took 0.21 seconds on map 5 and visited 6179 states.
15 moves were used: LULDDRDLUURUL
IDS took 2.98 seconds on map 6 and visited 7139 states.
34 moves were used: DRDDDRLLLLLLLLUUUUUUUUULRRRRRRR
IDS took 50.5 seconds on map 7 and visited 749203 states.
34 moves were used: RURRDDDLRUUUULLLRDRDRDLDLLUUDR
IDS took 0.24 seconds on map 8 and visited 6589 states.
14 moves were used: UURDLRRDRURDR
for map 9, we have big time to find solution.....
IDS took 86.83 seconds on map 10 and visited 236933 states.
46 moves were used: RRRRRDRULURULLLDLLURUULDRDLRRDRULURURDDRDLLLL

A*

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A* took 0.0 seconds on map 1 and visited 43 states.  
7 moves were used: DULRRLU  
A* took 0.0 seconds on map 2 and visited 21 states.  
6 moves were used: LDULRU  
A* took 0.0 seconds on map 3 and visited 80 states.  
13 moves were used: ULDDUUUURDDDD  
A* took 0.0 seconds on map 4 and visited 1 states.  
No Solution Found!  
A* took 0.01 seconds on map 5 and visited 890 states.  
15 moves were used: LULDDRLLUUVURUL  
A* took 1.4 seconds on map 6 and visited 5161 states.  
34 moves were used: DDDDDRRRLLLLLLUUUUUUUUURULRRRRRRR  
A* took 0.62 seconds on map 7 and visited 52672 states.  
34 moves were used: RURRDDDLD RUUUULLLRDRDRDDLLDLURLU  
A* took 0.02 seconds on map 8 and visited 410 states.  
14 moves were used: UURDLRRDRURDR  
for map 9, we have big time to find solution.....  
A* took 0.94 seconds on map 10 and visited 53245 states.  
46 moves were used: RRRRRDRULURULLDLLURUULDRDLRRDRULURURDDRDLLLL
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```
A* took 0.0 seconds on map 1 and visited 43 states.  
7 moves were used: DULRRLU  
A* took 0.0 seconds on map 2 and visited 21 states.  
6 moves were used: LDULRU  
A* took 0.0 seconds on map 3 and visited 80 states.  
13 moves were used: ULDDUUUURDDDD  
A* took 0.0 seconds on map 4 and visited 1 states.  
No Solution Found!  
A* took 0.01 seconds on map 5 and visited 890 states.  
15 moves were used: LULDDRLLUUVURUL  
A* took 0.04 seconds on map 6 and visited 5161 states.  
34 moves were used: DDDDDRRRLLLLLLUUUUUUUUURULRRRRRRR  
A* took 0.62 seconds on map 7 and visited 52672 states.  
34 moves were used: RURRDDDLD RUUUULLLRDRDRDDLLDLURLU  
A* took 0.01 seconds on map 8 and visited 410 states.  
14 moves were used: UURDLRRDRURDR  
for map 9, we have big time to find solution.....  
A* took 1.01 seconds on map 10 and visited 53245 states.  
46 moves were used: RRRRRDRULURULLDLLURUULDRDLRRDRULURURDDRDLLLL
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```
A* took 0.0 seconds on map 1 and visited 43 states.  
7 moves were used: DULRRLU  
A* took 0.0 seconds on map 2 and visited 21 states.  
6 moves were used: LDULRU  
A* took 0.0 seconds on map 3 and visited 80 states.  
13 moves were used: ULDDUUUURDDDD  
A* took 0.0 seconds on map 4 and visited 1 states.  
No Solution Found!  
A* took 0.01 seconds on map 5 and visited 890 states.  
15 moves were used: LULDDRLLUUVURUL  
A* took 0.05 seconds on map 6 and visited 5161 states.  
34 moves were used: DDDDDRRRLLLLLLUUUUUUUUURULRRRRRRR  
A* took 1.34 seconds on map 7 and visited 52672 states.  
34 moves were used: RURRDDDLD RUUUULLLRDRDRDDLLDLURLU  
A* took 0.0 seconds on map 8 and visited 410 states.  
14 moves were used: UURDLRRDRURDR  
for map 9, we have big time to find solution.....  
A* took 0.88 seconds on map 10 and visited 53245 states.  
46 moves were used: RRRRRDRULURULLDLLURUULDRDLRRDRULURURDDRDLLLL
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